

Dr. Tomáš Paus
Associate Editor
Human Brain Mapping

July 28, 2024

Dear Tomáš,

We respectfully submit our original manuscript entitled, “*Enhanced Detection of Age-Related and Cognitive Declines Using Automated Hippocampal-to-Ventricle Ratio in Alzheimer's Patients*” for publication in Human Brain Mapping as suggested by both Louis Lemieux and Nathan Sprang (editorial board member and associate editor, respectively).

As you are well aware, hippocampal atrophy is an early and important sign of neurodegeneration in Alzheimer's Disease. The high variability in the shape and size of the hippocampus make it difficult and time consuming to segment manually. Robust automatic measures of hippocampal neurodegeneration obtained from accurate and precise segmentations are needed.

To this end, we automated the Hippocampal-to-Ventricle Ratio (HVR) as a more robust biomarker compared to hippocampal volume alone for neurodegeneration of medial temporal lobe structures using three locally trained automatic segmentation methods (Multi-Atlas Label Fusion, MALF; Nonlinear Patch-Based Segmentation, NLPB; and a Convolutional Neural Network, CNN). We also compare to the widely used FreeSurfer algorithm. We extensively evaluated the four algorithms on large cohort of participants (n=1641) from the Alzheimer's Disease Neuroimaging Initiative (ADNI).

Key findings from our study include the superior performance of the CNN-based segmentation method in terms of accuracy, reliability, and clinical relevance compared to MALF and NLPB. We found the CNN-based segmentation was comparable to FreeSurfer. We also showed that, *particularly in cases of lower-quality segmentations*, HVR exhibits stronger correlations with age and cognitive measurements, and larger effect sizes between healthy controls and AD patients.

We believe that this work will be of interest to the broad audience of Human Brain Mapping as our results give insight into the importance of high-quality segmentation of the hippocampus for the study of neurodegeneration. In addition, the results can help users decide which is the best method to use for their application. The code for the analyses, together with our CNN model, will be made freely available to the public after publication.

We certify that this manuscript is submitted only to Human Brain Mapping, and it has not been, nor will be published in or submitted to any other journal. Please note that a very early preprint version of the work is available online through bioRxiv (BIORXIV/2024/588928). We have no affiliation, financial agreement or any other involvement with any company in connection with this research study. All authors have read the manuscript, have full access to the data and the right to publish with the data and take full responsibility for the data, analyses, interpretation and the conduct of research. On the following page, we include a list of suggested reviewers.

Thank you for considering our work for peer review in Human Brain Mapping.

Sincerely,

Sofia Fernández-Lozano, M.Sc.
Doctoral Candidate
Dept. Neurology & Neurosurgery
sofia.fernandezlozano@mail.mcgill.caD. Louis Collins, Ph.D.
James McGill Professor
Dept. Neurology & Neurosurgery and Biomedical Engineering
louis.collins@mcgill.ca

List of suggested reviewers:

- Josephine Barnes, University College London, Dementia Research Centre (j.barnes@ucl.ac.uk)
- Marie Chupin, ARAMIS team of the Brain and Spine Institute (marie.chupin@cnrs.fr)
- Paul Aljabar, Imperial College London (paul.aljabar@imperial.ac.uk)
- Marina Boccardi, Department of Psychiatry, LANVIE-Laboratory of Neuroimaging of Aging, University of Geneva, (mboccardifbf@googlemail.com)
- Eleanor Maguire, Wellcome Centre for Human Neuroimaging, UCL, (e.maguire@ucl.ac.uk)
- Ali Khan, assistant professor, University of Western Ontario (ali.khan@uwo.ca)
- Sean Nestor, assistant professor, Neurosciences and Clinical Translation, Sunnybrook, (sean.nestor@sunnybrook.ca)
- Bruce Fischl, MGH, Harvard, (bfischl@mgh.harvard.edu)
- Hongzhi Wang, IBM Almaden Research Center (hongzhiw@mail.med.upenn.edu)